ACTIVITY GUIDE

Network Playground

A Packet Tracer-style sandbox. Drag devices onto the canvas, click to cable them, and the playground auto-configures the addressing. No commands — the focus is the topology.

Building a network

1. **Add devices:** drag a PC, Switch or Router from the palette onto the canvas (or click one to drop it in the middle).
2. **Move:** in Build mode, drag any device to reposition it. Use **Auto-arrange** to tidy into tiers.
3. **Wire:** click the **Wire** tool, then click one device and then a second to cable them together. Ports are chosen for you.
4. **Delete:** click the **Delete** tool, then click a device or a cable to remove it.
5. Zoom with the `+ − ⤢` buttons or Ctrl-scroll — the canvas zooms without changing the page size.

How auto-configuration works

- Each **broadcast domain** (devices joined without crossing a router) becomes its own `192.168.x.0/24` subnet.
- PCs receive host addresses (`.10`, `.11`…); a router interface on that subnet becomes the gateway (`.1`).
- A PC with no cable shows **no IP** — it isn't connected to anything yet.

Testing with ping

1. Pick a **From** and **To** PC in the Ping test bar and click **Ping →**.
2. The log explains the result hop by hop and a packet animates along the path.

Challenges

- Build two PCs on one switch and make them ping (same subnet).
- Add a second switch and a router so two subnets can talk through the gateway.
- Remove the router — watch the ping fail with *"no route"* and explain why.
- Build a three-router network and test reachability across it.

Remember: switches are Layer 2 (they don't get an IP); only routers separate subnets and provide gateways.

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